

BENJAMIN BLACK

Postdoctoral researcher with a professional background in conservation, specialising in socio-environmental systems modelling to support evidence-based policy and planning. Passionate about advancing open science, communicating research beyond academia, and mentoring others in modelling, programmatic data analysis, and visualisation. Experienced in managing small and large teams, especially in cross-cultural and interdisciplinary settings, with a strong commitment to collaborative and inclusive research.



EDUCATION

Oct. 2020
|
Jun. 2025

Doctor of Civil Engineering

ETH Zürich, chair of Planning of Landscape Urban Systems (PLUS)

📍 Zürich, Switzerland

Thesis: Identifying effective area-based conservation strategies under uncertainties

Awards: Nominated for the ETH medal for outstanding doctoral theses

Sep. 2018
|
Jun. 2020

MSc in Environmental Sciences, Policy and Management (MESPOM)

Central European University, University of Manchester, Lund University, University of the Aegean

📍 Hungary, Greece, UK

Thesis: Assessing the ecological effectiveness of protected areas in Cambodia: a quasi-experimental counterfactual estimation of avoided deforestation

Awards: Outstanding academic achievement (highest GPA in cohort: 4.00/4.00)

Sep. 2010
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Jun. 2014

BSc with 1st Class Honours in Biological Sciences

Edinburgh Napier University

📍 Edinburgh, Scotland

Thesis: Simulating the future potential of degraded Mangrove plots using field data from Kenya

Awards: University medal (highest average mark on degree route); Lawrence Ho student prize; Dennis Aw award.



PROFESSIONAL EXPERIENCE

May. 2025
|
Jul. 2025

Freelance Research Consultant

Prognos AG

📍 Zürich, Switzerland

Performed Land Use and Land Cover change modelling of Shared Socioeconomic Pathways for Switzerland as part of project funded by the Swiss National Centre for Climate Services

Jul. 2019
|
Aug. 2019

Biodiversity and SMART Intern

The Wildlife Conservation Society (WCS)

📍 Phnom Penh, Cambodia

Developed a programmatic routine to expedite the collection of external data from the IUCN, EDGE ranking and CITES to produce protected area species inventories in line with Darwin Core standards. Assisted in the training of new community biodiversity rangers in the process of SMART data collection in the field.

Jan. 2018
|
Aug. 2018

Student Affairs Manager

The School for Field Studies Center for Conservation and Development Studies in the Lower Mekong

📍 Siem Reap, Cambodia

Coordinated safety and risk management as well as logistics for all centre activities including multi day field trips to remote locations. Served as the medical point person, facilitating first aid training for staff, while also supporting student well-being and cultural adaptation in a pastoral role.



CONTACT INFO

👤 Postdoctoral researcher

🏛️ ETH Planning of Landscape and Urban systems

📍 Zurich, Switzerland

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🌀 blenback

✂️ Blen_Back

🌐 <https://www.researchgate.net/profile/Benjamin-Black-5>

TECHNICAL SKILLS

Advanced: R (Shiny, Quarto), Dinamica EGO, Netica and MS Office

Intermediate: ArcGIS, QGIS, CSS and Git

Basic: Python, HTML, Docker, JavaScript, Bash and Slurm

LANGUAGES

English: Native

German: B2

- April. 2015
June. 2017

Project Manager & Volunteer Co-ordinator
Elephant Livelihood Initiatives and Environment (E.L.I.E)
📍 Mondulkiri, Cambodia

Managed a diverse team (Bunong Indigenous minority, and Khmer) of 48 staff. Oversaw the operation of an ecotourism project receiving 4000+ visitors a year. Facilitated ongoing capacity building of staff and development of the volunteer program. Compiled budgets, managed cash flow, and created detailed financial and organizational reports.
- Oct. 2015
Dec. 2015

Independent Consultant Srae Y Community ecotourism project
World Wide Fund for Nature (WWF)
📍 Mondulkiri, Cambodia

Undertook site visits to assess the status of the project and prepared a report, detailing strategies to make the project operational and sustainable. Planned and delivered a series of training sessions to project staff and stakeholders from other communities engaged in ecotourism.
- May 2013
Apr. 2015

Volunteer Development Officer & Conservation Support Assistant
The Conservation Volunteers (TCV)
📍 Central Scotland

Planned, developed and delivered the Ochils Landscape Partnerships volunteering program and two TCV community health projects. Engaged with, and ensured the safety and wellbeing of a diverse cross section of volunteers, many with physical or mental disabilities as well as young adults/children.



TEACHING, TRAINING & WORKSHOP DELIVERY

- Sep. 2024

Reproducible Research with R and Quarto: Workflows for data, projects and publications
Three hour workshop as part of Landscape 2024 conference
📍 Berlin, Germany
- Feb. 2021
Jul. 2024

Participatory Environmental Modelling: MSc course at ETH Zürich
Designed and taught an MSc-level course integrating students and practitioners to develop simulation models for real-world environmental decision-making.
📍 Zürich, Switzerland

 Website



GRANTS & FUNDING

- Dec. 2024

Future development of biodiversity and ecosystem services as a basis for the revision of the sectoral plan for biodiversity
Canton of Bern Department of Agriculture and Nature
📍 Bern, Switzerland

Project in collaboration with the Wyss Academy for Nature to generate policy-relevant evidence supporting the canton's revision of the sectoral biodiversity plan.
- Jun. 2023

Nature-positive Scenarios for Environmental Transitions Peru (NASCENT-PERU)
Swiss Re Foundation
📍 Zürich, Switzerland

Funded project under the SwissRE Biodiversity and Ecosystem Services Scenarios Modelling Initiative

 Website



PUBLICATIONS

- In Review

The PAVE pathway: how to talk about models with people who did(n't) build them
Bell, A., Troost, C., Filatova, T., Vincenot, C., Qin, C., Fiedler, S., Lemmen, C., Lynch, I., Pierce, M., Edmonds, B., McIntire, E., Wang, H., Koralewski, T., Grant, W., Wang, M., Black, B., Grimm, V.
Bioscience
- In Review

Where will the change come from? Identifying decision-relevant factors in future Land Use and Cover Change ({LUCC}) scenarios to support planning under uncertainty
Roman, O., Adey, B., Black, B. and Zeindl, E.
Futures

In Review	Temporal Archetypes of Nature's Contributions to People for Sustainable Landscape Development Wicki, S., Black, B., Külling, N., Kurmann, M., Wang, J., Lehmann, A., Gret-Regamey, A. <i>Regional Environmental Change</i>	
In Review	Identifying robust conservation strategies to secure ecosystem service provision under uncertainties Black, B. and Adde, A. and Külling, N. and Büth, C. and Kurmann, M. and Lehmann, A. and Altermatt, F. and Guisan, A. and Gret-Regamey, A. <i>Journal of Environmental Management</i>	
2024	Broadening the horizon in land use change modelling: Normative scenarios for nature positive futures in Switzerland Black, B. and Adde, A. and Farinotti, D. and Guisan, A. and Külling, N. and Kurmann, M. and Martin, C. and Mayer, P. and Rabe, S and Streit, J. and Zekollari, H. and Grêt-Regamey, A. <i>Regional Environmental Change</i>	Read Data Code
2023	Archetypes of social-ecological-technological systems for managing ecological infrastructure Wicki, S. and Black, B. and Kurmann, M. and Grêt-Regamey, A. <i>Environmental Research Letters</i>	Read
2023	Re-considering the status quo: Improving calibration of land use change models through validation of transition potential predictions Black, B. and van Strien, M. J. and Adde, A. and Grêt-Regamey, A. <i>Environmental Modelling & Software</i>	Read Code
2022	Counterfactual assessment of protected area avoided deforestation in Cambodia: Trends in effectiveness, spillover effects and the influence of establishment date Black, B. and Anthony, B. P. <i>Global Ecology and Conservation</i>	Read Data



PRESENTATIONS

Jun. 2025	A “Scope, Simulation and Story” approach to identify future spatial development scenarios Computational Urban Planning and Urban Management Conference 2025 📍 University College London	Slides
Sep. 2024	Negotiating scenarios of a desirable ecological infrastructure for Switzerland 4th Swiss Landscape Congress 📍 Tramelan, Switzerland	Slides
Jun. 2024	Normative scenarios of landscape change for Nature-Positive futures in Peru World Biodiversity Forum 📍 Davos, Switzerland Exploratory modelling of stochastic land use cover change (LUCC) future scenarios for spatial planning 10th Annual Conference of the Society for Decision Making Under Deep Uncertainty (DMDU 2023) 📍 Delft, Netherlands	Slides
Nov. 2023	Broadening the scope of future visions for nature positive futures in Peru Swiss Re Institute Resilience Summit 2023 📍 Zürich, Switzerland	Slides
Sep. 2022	Characterising non-stationarity in predictive models of land use in Swiss mountain parks to inform scenarios for deliberative transformation International Mountain conference 2022 📍 Innsbruck, Austria	Slides

Jul. 2022



ValPar.CH: Integrating land use change, Ecosystem Service and Biodiversity modelling to simulate pathways for a functioning Ecological Infrastructure for Switzerland

11th International Environmental Modelling and Software Society conference



Brussels, Belgium

 Slides

Jul. 2022



Combining filter and embedded approaches to improve variable selection in land use change Cellular Automata models

11th International Environmental Modelling and Software Society conference



Brussels, Belgium

 Slides

Jun. 2019



Ex-ante identification of policy interventions to secure a functioning ecological infrastructure: A participatory Bayesian Network approach in Switzerland

3rd Ecosystem Services Partnership Europe Regional conference



Tartu, Estonia

 Slides